



Adele ring

In mathematics, the **adele ring** of a global field (also **adelic ring**, **ring of adeles** or **ring of adèles**^[1]) is a central object of class field theory, a branch of algebraic number theory. It is the restricted product of all the completions of the global field and is an example of a self-dual topological ring.

An adele derives from a particular kind of **idele**. "Idele" derives from the French "idèle" and was coined by the French mathematician Claude Chevalley. The word stands for 'ideal element' (abbreviated: id.el.). **Adele** (French: "adèle") stands for 'additive idele' (that is, additive ideal element).

The ring of adeles allows one to describe the Artin reciprocity law, which is a generalisation of quadratic reciprocity, and other reciprocity laws over finite fields. In addition, it is a classical theorem from Weil that *G*-bundles on an algebraic curve over a finite field can be described in terms of adeles for a reductive group *G*. Adeles are also connected with the adelic algebraic groups and adelic curves.

The study of geometry of numbers over the ring of adeles of a number field is called **adelic geometry**.

Definition

Let *K* be a global field (a finite extension of **Q** or the function field of a curve *X/F_q* over a finite field). The **adele ring** of *K* is the subring

$$\mathbf{A}_K = \prod (K_\nu, \mathcal{O}_\nu) \subseteq \prod K_\nu$$

consisting of the tuples (*a_ν*) where *a_ν* lies in the subring *ℳ_ν* ⊂ *K_ν* for all but finitely many places *ν*. Here the index *ν* ranges over all valuations of the global field *K*, *K_ν* is the completion at that valuation and *ℳ_ν* the corresponding valuation ring.^[2]

Motivation

The ring of adeles solves the technical problem of "doing analysis on the rational numbers **Q**." The classical solution was to pass to the standard metric completion **R** and use analytic techniques there. But, as was learned later on, there are many more absolute values other than the Euclidean distance, one for each prime number *p* ∈ **Z**, as classified by Ostrowski's theorem. The Euclidean absolute value, denoted | · |_∞, is only one among many others, | · |_{*p*}, but the ring of adeles makes it possible to comprehend and *use all of the valuations at once*. This has the advantage of enabling analytic techniques while also retaining information about the primes, since their structure is embedded by the restricted infinite product.

The purpose of the adele ring is to look at all completions of *K* at once. The adele ring is defined with the restricted product, rather than the Cartesian product. There are two reasons for this:

- For each element of *K* the valuations are zero for almost all places, i.e., for all places except a finite number. So, the global field can be embedded in the restricted product.
- The restricted product is a locally compact space, while the Cartesian product is not. Therefore, there cannot be any application of harmonic analysis to the Cartesian product. This is because local compactness ensures the existence (and uniqueness) of Haar measure, a crucial tool in analysis on groups in general.

Why the restricted product?

The restricted infinite product is a required technical condition for giving the number field **Q** a lattice structure inside of **A_Q**, making it possible to build a theory of Fourier analysis (cf. Harmonic analysis) in the adelic setting. This is analogous to the situation in algebraic number theory where the ring of integers of an algebraic number field embeds

$$\mathcal{O}_K \hookrightarrow K$$

as a lattice. With the power of a new theory of Fourier analysis, Tate was able to prove a special class of L-functions and the Dedekind zeta functions were meromorphic on the complex plane. Another natural reason for why this technical condition holds can be seen by constructing the ring of adeles as a tensor product of rings. If defining the ring of integral adeles $\mathbf{A}_{\mathbf{Z}}$ as the ring

$$\mathbf{A}_{\mathbf{Z}} = \mathbf{R} \times \hat{\mathbf{Z}} = \mathbf{R} \times \prod_p \mathbf{Z}_p,$$

then the ring of adeles can be equivalently defined as

$$\begin{aligned} \mathbf{A}_{\mathbf{Q}} &= \mathbf{Q} \otimes_{\mathbf{Z}} \mathbf{A}_{\mathbf{Z}} \\ &= \mathbf{Q} \otimes_{\mathbf{Z}} \left(\mathbf{R} \times \prod_p \mathbf{Z}_p \right). \end{aligned}$$

The restricted product structure becomes transparent after looking at explicit elements in this ring. The image of an element $b/c \otimes (r, (a_p)) \in \mathbf{A}_{\mathbf{Q}}$ inside of the unrestricted product $\mathbf{R} \times \prod_p \mathbf{Q}_p$ is the element

$$\left(\frac{br}{c}, \left(\frac{ba_p}{c} \right) \right).$$

The factor ba_p/c lies in $\mathbf{Z}_p$ whenever p is not a prime factor of c , which is the case for all but finitely many primes p .^[3]

Origin of the name

The term "idele" (French: *idèle*) is an invention of the French mathematician Claude Chevalley (1909–1984) and stands for "ideal element" (abbreviated: id.el.). The term "adele" (French: *adèle*) stands for additive idele. Thus, an adele is an additive ideal element.

Examples

Ring of adeles for the rational numbers

The rationals $K = \mathbf{Q}$ have a valuation for every prime number p , with $(K_{\nu}, \mathcal{O}_{\nu}) = (\mathbf{Q}_p, \mathbf{Z}_p)$, and one infinite valuation ∞ with $\mathbf{Q}_{\infty} = \mathbf{R}$. Thus an element of

$$\mathbf{A}_{\mathbf{Q}} = \mathbf{R} \times \prod_p (\mathbf{Q}_p, \mathbf{Z}_p)$$

is a real number along with a p -adic rational for each p of which all but finitely many are p -adic integers.

Ring of adeles for the function field of the projective line

Secondly, take the function field $K = \mathbf{F}_q(\mathbf{P}^1) = \mathbf{F}_q(t)$ of the projective line over a finite field. Its valuations correspond to points x of $X = \mathbf{P}^1$, i.e. maps over $\mathbf{Spec} \mathbf{F}_q$

$$x : \operatorname{Spec} \mathbf{F}_{q^n} \longrightarrow \mathbf{P}^1.$$

For instance, there are $q+1$ points of the form $\operatorname{Spec} \mathbf{F}_q \longrightarrow \mathbf{P}^1$. In this case $\mathcal{O}_\nu = \widehat{\mathcal{O}}_{X,x}$ is the completed stalk of the structure sheaf at x (i.e. functions on a formal neighbourhood of x) and $K_\nu = K_{X,x}$ is its fraction field. Thus

$$\mathbf{A}_{\mathbf{F}_q(\mathbf{P}^1)} = \prod_{x \in X} (\mathcal{K}_{X,x}, \widehat{\mathcal{O}}_{X,x}).$$

The same holds for any smooth proper curve $X/\mathbf{F}_q$ over a finite field, the restricted product being over all points of $x \in X$.

Related notions

The group of units in the adèle ring is called the **idele group**

$$I_K = \mathbf{A}_K^\times.$$

The quotient of the ideles by the subgroup $K^\times \subseteq I_K$ is called the **idele class group**

$$C_K = I_K / K^\times.$$

The **integral adeles** are the subring

$$\mathbf{O}_K = \prod O_\nu \subseteq \mathbf{A}_K.$$

Applications

Stating Artin reciprocity

The Artin reciprocity law says that for a global field K ,

$$\widehat{C_K} = \widehat{\mathbf{A}_K^\times / K^\times} \simeq \operatorname{Gal}(K^{\text{ab}}/K)$$

where K^{ab} is the maximal abelian algebraic extension of K and $\widehat{(\dots)}$ means the profinite completion of the group.

Giving adelic formulation of Picard group of a curve

If $X/\mathbf{F}_q$ is a smooth proper curve then its Picard group is^[4]

$$\operatorname{Pic}(X) = K^\times \backslash \mathbf{A}_X^\times / \mathbf{O}_X^\times$$

and its divisor group is $\operatorname{Div}(X) = \mathbf{A}_X^\times / \mathbf{O}_X^\times$. Similarly, if G is a semisimple algebraic group (e.g. SL_n , it also holds for GL_n) then Weil uniformisation says that^[5]

$$\operatorname{Bun}_G(X) = G(K) \backslash G(\mathbf{A}_X) / G(\mathbf{O}_X).$$

Applying this to $G = \mathbf{G}_m$ gives the result on the Picard group.

Tate's thesis

There is a topology on $\mathbf{A}_K$ for which the quotient $\mathbf{A}_K/K$ is compact, allowing one to do harmonic analysis on it. John Tate in his thesis "Fourier analysis in number fields and Hecke Zeta functions"^[6] proved results about Dirichlet L-functions using Fourier analysis on the adèle ring and the idele group. Therefore, the adèle ring and the idele group have been applied to study the Riemann zeta function and more general zeta functions and the L-functions.

Proving Serre duality on a smooth curve

If X is a smooth proper curve *over the complex numbers*, one can define the adèles of its function field $\mathbf{C}(X)$ exactly as the finite fields case. John Tate proved^[7] that Serre duality on X

$$H^1(X, \mathcal{L}) \simeq H^0(X, \Omega_X \otimes \mathcal{L}^{-1})^*$$

can be deduced by working with this adèle ring $\mathbf{A}_{\mathbf{C}(X)}$. Here L is a line bundle on X .

Notation and basic definitions

Global fields

Throughout this article, K is a global field, meaning it is either a number field (a finite extension of $\mathbb{Q}$) or a global function field (a finite extension of $\mathbb{F}_{p^r}(t)$ for p prime and $r \in \mathbb{N}$). By definition a finite extension of a global field is itself a global field.

Valuations

For a valuation v of K it can be written K_v for the completion of K with respect to v . If v is discrete it can be written $\mathcal{O}_v$ for the valuation ring of K_v and $\mathfrak{m}_v$ for the maximal ideal of $\mathcal{O}_v$. If this is a principal ideal denoting the uniformising element by π_v . A non-Archimedean valuation is written as $v < \infty$ or $v \nmid \infty$ and an Archimedean valuation as $v|\infty$. Then assume all valuations to be non-trivial.

There is a one-to-one identification of valuations and absolute values. Fix a constant $C > 1$, the valuation v is assigned the absolute value $|\cdot|_v$, defined as:

$$\forall x \in K : \quad |x|_v := \begin{cases} C^{-v(x)} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Conversely, the absolute value $|\cdot|$ is assigned the valuation $v_{|\cdot|}$, defined as:

$$\forall x \in K^\times : \quad v_{|\cdot|}(x) := -\log_C(|x|).$$

A place of K is a representative of an equivalence class of valuations (or absolute values) of K . Places corresponding to non-Archimedean valuations are called *finite*, whereas places corresponding to Archimedean valuations are called *infinite*. Infinite places of a global field form a finite set, which is denoted by P_∞ .

Define $\hat{\mathcal{O}} := \prod_{v < \infty} \mathcal{O}_v$ and let $\hat{\mathcal{O}}^\times$ be its group of units. Then $\hat{\mathcal{O}}^\times = \prod_{v < \infty} \mathcal{O}_v^\times$.

Finite extensions

Let L/K be a finite extension of the global field K . Let w be a place of L and v a place of K . If the absolute value $|\cdot|_w$ restricted to K is in the equivalence class of v , then w lies above v , which is denoted by $w|v$, and defined as:

$$L_v := \prod_{w|v} L_w,$$

$$\widetilde{O}_v := \prod_{w|v} O_w.$$

(Note that both products are finite.)

If $w|v$, K_v can be embedded in L_w . Therefore, K_v is embedded diagonally in L_v . With this embedding L_v is a commutative algebra over K_v with degree

$$\sum_{w|v} [L_w : K_v] = [L : K].$$

The adèle ring

The set of **finite adeles of a global field K** , denoted $\mathbb{A}_{K,\text{fin}}$, is defined as the restricted product of K_v with respect to the O_v :

$$\mathbb{A}_{K,\text{fin}} := \prod'_{v<\infty} K_v = \left\{ (x_v)_v \in \prod_{v<\infty} K_v \mid x_v \in O_v \text{ for almost all } v \right\}.$$

It is equipped with the restricted product topology, the topology generated by restricted open rectangles, which have the following form:

$$U = \prod_{v \in E} U_v \times \prod_{v \notin E} O_v \subset \prod'_{v<\infty} K_v,$$

where E is a finite set of (finite) places and $U_v \subset K_v$ are open. With component-wise addition and multiplication $\mathbb{A}_{K,\text{fin}}$ is also a ring.

The **adele ring of a global field K** is defined as the product of $\mathbb{A}_{K,\text{fin}}$ with the product of the completions of K at its infinite places. The number of infinite places is finite and the completions are either $\mathbb{R}$ or $\mathbb{C}$. In short:

$$\mathbb{A}_K := \mathbb{A}_{K,\text{fin}} \times \prod_{v|\infty} K_v = \prod'_{v<\infty} K_v \times \prod_{v|\infty} K_v.$$

With addition and multiplication defined as component-wise the adele ring is a ring. The elements of the adele ring are called **adeles of K** . In the following, it is written as

$$\mathbb{A}_K = \prod'_v K_v,$$

although this is generally not a restricted product.

Remark. Global function fields do not have any infinite places and therefore the finite adele ring equals the adele ring.

Lemma. There is a natural embedding of K into $\mathbb{A}_K$ given by the diagonal map:
 $a \mapsto (a, a, \dots).$

Proof. If $a \in K$, then $a \in O_v^\times$ for almost all v . This shows the map is well-defined. It is also injective because the embedding of K in K_v is injective for all v .

Remark. By identifying K with its image under the diagonal map it is regarded as a subring of $\mathbb{A}_K$. The elements of K are called the **principal adeles** of $\mathbb{A}_K$.

Definition. Let S be a set of places of K . Define the **set of the S -adeles of K** as

$$\mathbb{A}_{K,S} := \prod'_{v \in S} K_v.$$

Furthermore, if

$$\mathbb{A}_K^S := \prod'_{v \notin S} K_v$$

the result is: $\mathbb{A}_K = \mathbb{A}_{K,S} \times \mathbb{A}_K^S$.

The adele ring of rationals

By Ostrowski's theorem the places of $\mathbb{Q}$ are $\{p \in \mathbb{N} : p \text{ prime}\} \cup \{\infty\}$, it is possible to identify a prime p with the equivalence class of the p -adic absolute value and ∞ with the equivalence class of the absolute value $|\cdot|_\infty$ defined as:

$$\forall x \in \mathbb{Q} : \quad |x|_\infty := \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

The completion of $\mathbb{Q}$ with respect to the place p is $\mathbb{Q}_p$ with valuation ring $\mathbb{Z}_p$. For the place ∞ the completion is $\mathbb{R}$. Thus:

$$\begin{aligned} \mathbb{A}_{\mathbb{Q}, \text{fin}} &= \prod'_{p < \infty} \mathbb{Q}_p \\ \mathbb{A}_{\mathbb{Q}} &= \left(\prod'_{p < \infty} \mathbb{Q}_p \right) \times \mathbb{R} \end{aligned}$$

Or for short

$$\mathbb{A}_{\mathbb{Q}} = \prod'_{p \leq \infty} \mathbb{Q}_p, \quad \mathbb{Q}_{\infty} := \mathbb{R}.$$

the difference between restricted and unrestricted product topology can be illustrated using a sequence in $\mathbb{A}_{\mathbb{Q}}$:

Lemma. Consider the following sequence in $\mathbb{A}_{\mathbb{Q}}$:

$$\begin{aligned} x_1 &= \left(\frac{1}{2}, 1, 1, \dots \right) \\ x_2 &= \left(1, \frac{1}{3}, 1, \dots \right) \\ x_3 &= \left(1, 1, \frac{1}{5}, 1, \dots \right) \\ x_4 &= \left(1, 1, 1, \frac{1}{7}, 1, \dots \right) \\ &\vdots \end{aligned}$$

In the product topology this converges to $(1, 1, \dots)$, but it does not converge at all in the restricted product topology.

Proof. In product topology convergence corresponds to the convergence in each coordinate, which is trivial because the sequences become stationary. The sequence doesn't converge in restricted product topology. For each adele $a = (a_p)_p \in \mathbb{A}_{\mathbb{Q}}$ and for each restricted open rectangle $U = \prod_{p \in E} U_p \times \prod_{p \notin E} \mathbb{Z}_p$, it has: $\frac{1}{p} - a_p \notin \mathbb{Z}_p$ for $a_p \in \mathbb{Z}_p$ and therefore $\frac{1}{p} - a_p \notin \mathbb{Z}_p$ for all $p \notin F$. As a result $x_n - a \notin U$ for almost all $n \in \mathbb{N}$. In this consideration, E and F are finite subsets of the set of all places.

Alternative definition for number fields

Definition (profinite integers). The **profinite integers** are defined as the profinite completion of the rings $\mathbb{Z}/n\mathbb{Z}$ with the partial order $n \geq m \Leftrightarrow m|n$, i.e.,

$$\hat{\mathbb{Z}} := \varprojlim_n \mathbb{Z}/n\mathbb{Z},$$

Lemma. $\hat{\mathbb{Z}} \cong \prod_p \mathbb{Z}_p$.

Proof. This follows from the Chinese Remainder Theorem.

Lemma. $\mathbb{A}_{\mathbb{Q}, \text{fin}} = \hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}$.

Proof. Use the universal property of the tensor product. Define a $\mathbb{Z}$ -bilinear function

$$\begin{cases} \Psi : \widehat{\mathbb{Z}} \times \mathbb{Q} \rightarrow \mathbb{A}_{\mathbb{Q}, \text{fin}} \\ ((a_p)_p, q) \mapsto (a_p q)_p \end{cases}$$

This is well-defined because for a given $q = \frac{m}{n} \in \mathbb{Q}$ with m, n co-prime there are only finitely many primes dividing n . Let M be another $\mathbb{Z}$ -module with a $\mathbb{Z}$ -bilinear map $\Phi : \widehat{\mathbb{Z}} \times \mathbb{Q} \rightarrow M$. It must be the case that Φ factors through Ψ uniquely, i.e., there exists a unique $\mathbb{Z}$ -linear map $\tilde{\Phi} : \mathbb{A}_{\mathbb{Q}, \text{fin}} \rightarrow M$ such that $\Phi = \tilde{\Phi} \circ \Psi$. $\tilde{\Phi}$ can be defined as follows: for a given $(u_p)_p$ there exist $u \in \mathbb{N}$ and $(v_p)_p \in \widehat{\mathbb{Z}}$ such that $u_p = \frac{1}{u} \cdot v_p$ for all p . Define $\tilde{\Phi}((u_p)_p) := \Phi((v_p)_p, \frac{1}{u})$. One can show $\tilde{\Phi}$ is well-defined, $\mathbb{Z}$ -linear, satisfies $\Phi = \tilde{\Phi} \circ \Psi$ and is unique with these properties.

Corollary. Define $\mathbb{A}_{\mathbb{Z}} := \widehat{\mathbb{Z}} \times \mathbb{R}$. This results in an algebraic isomorphism $\mathbb{A}_{\mathbb{Q}} \cong \mathbb{A}_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}$.

Proof. $\mathbb{A}_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q} = (\widehat{\mathbb{Z}} \times \mathbb{R}) \otimes_{\mathbb{Z}} \mathbb{Q} \cong (\widehat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}) \times (\mathbb{R} \otimes_{\mathbb{Z}} \mathbb{Q}) \cong (\widehat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}) \times \mathbb{R} = \mathbb{A}_{\mathbb{Q}, \text{fin}} \times \mathbb{R} = \mathbb{A}_{\mathbb{Q}}$.

Lemma. For a number field K , $\mathbb{A}_K = \mathbb{A}_{\mathbb{Q}} \otimes_{\mathbb{Q}} K$.

Remark. Using $\mathbb{A}_{\mathbb{Q}} \otimes_{\mathbb{Q}} K \cong \mathbb{A}_{\mathbb{Q}} \oplus \dots \oplus \mathbb{A}_{\mathbb{Q}}$, where there are $[K : \mathbb{Q}]$ summands, give the right side receives the product topology and transport this topology via the isomorphism onto $\mathbb{A}_{\mathbb{Q}} \otimes_{\mathbb{Q}} K$.

The adèle ring of a finite extension

If L/K be a finite extension, then L is a global field. Thus $\mathbb{A}_L$ is defined, and $\mathbb{A}_L = \prod'_v L_v$. $\mathbb{A}_K$ can be identified with a subgroup of $\mathbb{A}_L$. Map $a = (a_v)_v \in \mathbb{A}_K$ to $a' = (a'_w)_w \in \mathbb{A}_L$ where $a'_w = a_v \in K_v \subset L_w$ for $w|v$. Then $a = (a_w)_w \in \mathbb{A}_L$ is in the subgroup $\mathbb{A}_K$, if $a_w \in K_v$ for $w|v$ and $a_w = a_{w'}$ for all w, w' lying above the same place v of K .

Lemma. If L/K is a finite extension, then $\mathbb{A}_L \cong \mathbb{A}_K \otimes_K L$ both algebraically and topologically.

With the help of this isomorphism, the inclusion $\mathbb{A}_K \subset \mathbb{A}_L$ is given by

$$\begin{cases} \mathbb{A}_K \rightarrow \mathbb{A}_L \\ \alpha \mapsto \alpha \otimes_K 1 \end{cases}$$

Furthermore, the principal adeles in $\mathbb{A}_K$ can be identified with a subgroup of principal adeles in $\mathbb{A}_L$ via the map

$$\begin{cases} K \rightarrow (K \otimes_K L) \cong L \\ \alpha \mapsto 1 \otimes_K \alpha \end{cases}$$

Proof.^[8] Let $\omega_1, \dots, \omega_n$ be a basis of L over K . Then for almost all v ,

$$\widetilde{O}_v \cong O_v \omega_1 \oplus \dots \oplus O_v \omega_n.$$

Furthermore, there are the following isomorphisms:

$$K_v \omega_1 \oplus \cdots \oplus K_v \omega_n \cong K_v \otimes_K L \cong L_v = \prod_{w|v} L_w$$

For the second use the map:

$$\begin{cases} K_v \otimes_K L \rightarrow L_v \\ \alpha_v \otimes a \mapsto (\alpha_v \cdot (\tau_w(a)))_w \end{cases}$$

in which $\tau_w : L \rightarrow L_w$ is the canonical embedding and $w|v$. The restricted product is taken on both sides with respect to $\widetilde{\mathcal{O}}_v$:

$$\begin{aligned} \mathbb{A}_K \otimes_K L &= \left(\prod'_v K_v \right) \otimes_K L \\ &\cong \prod'_v (K_v \omega_1 \oplus \cdots \oplus K_v \omega_n) \\ &\cong \prod'_v (K_v \otimes_K L) \\ &\cong \prod'_v L_v \\ &= \mathbb{A}_L \end{aligned}$$

Corollary. As additive groups $\mathbb{A}_L \cong \mathbb{A}_K \oplus \cdots \oplus \mathbb{A}_K$, where the right side has $[L : K]$ summands.

The set of principal adeles in $\mathbb{A}_L$ is identified with the set $K \oplus \cdots \oplus K$, where the left side has $[L : K]$ summands and K is considered as a subset of $\mathbb{A}_K$.

The adèle ring of vector-spaces and algebras

Lemma. Suppose $P \supset P_\infty$ is a finite set of places of K and define

$$\mathbb{A}_K(P) := \prod_{v \in P} K_v \times \prod_{v \notin P} \mathcal{O}_v.$$

Equip $\mathbb{A}_K(P)$ with the product topology and define addition and multiplication component-wise. Then $\mathbb{A}_K(P)$ is a locally compact topological ring.

Remark. If P' is another finite set of places of K containing P then $\mathbb{A}_K(P)$ is an open subring of $\mathbb{A}_K(P')$.

Now, an alternative characterisation of the adèle ring can be presented. The adèle ring is the union of all sets $\mathbb{A}_K(P)$:

$$\mathbb{A}_K = \bigcup_{P \supset P_\infty, |P| < \infty} \mathbb{A}_K(P).$$

Equivalently $\mathbb{A}_K$ is the set of all $\mathbf{x} = (x_v)_v$ so that $|x_v|_v \leq 1$ for almost all $v < \infty$. The topology of $\mathbb{A}_K$ is induced by the requirement that all $\mathbb{A}_K(P)$ be open subrings of $\mathbb{A}_K$. Thus, $\mathbb{A}_K$ is a locally compact topological ring.

Fix a place v of K . Let P be a finite set of places of K , containing v and P_∞ . Define

$$\mathbb{A}'_K(P, v) := \prod_{w \in P \setminus \{v\}} K_w \times \prod_{w \notin P} O_w.$$

Then:

$$\mathbb{A}_K(P) \cong K_v \times \mathbb{A}'_K(P, v).$$

Furthermore, define

$$\mathbb{A}'_K(v) := \bigcup_{P \supset P_\infty \cup \{v\}} \mathbb{A}'_K(P, v),$$

where P runs through all finite sets containing $P_\infty \cup \{v\}$. Then:

$$\mathbb{A}_K \cong K_v \times \mathbb{A}'_K(v),$$

via the map $(a_w)_w \mapsto (a_v, (a_w)_{w \neq v})$. The entire procedure above holds with a finite subset $\tilde{P}$ instead of $\{v\}$.

By construction of $\mathbb{A}'_K(v)$, there is a natural embedding: $K_v \hookrightarrow \mathbb{A}_K$. Furthermore, there exists a natural projection $\mathbb{A}_K \rightarrow K_v$.

The adèle ring of a vector-space

Let E be a finite dimensional vector-space over K and $\{\omega_1, \dots, \omega_n\}$ a basis for E over K . For each place v of K :

$$\begin{aligned} E_v &:= E \otimes_K K_v \cong K_v \omega_1 \oplus \dots \oplus K_v \omega_n \\ \widetilde{O}_v &:= O_v \omega_1 \oplus \dots \oplus O_v \omega_n \end{aligned}$$

The adèle ring of E is defined as

$$\mathbb{A}_E := \prod'_v E_v.$$

This definition is based on the alternative description of the adèle ring as a tensor product equipped with the same topology that was defined when giving an alternate definition of adèle ring for number fields. Next, $\mathbb{A}_E$ is equipped with the restricted product topology. Then $\mathbb{A}_E = E \otimes_K \mathbb{A}_K$ and E is embedded in $\mathbb{A}_E$ naturally via the map $e \mapsto e \otimes 1$.

An alternative definition of the topology on $\mathbb{A}_E$ can be provided. Consider all linear maps: $E \rightarrow K$. Using the natural embeddings $E \rightarrow \mathbb{A}_E$ and $K \rightarrow \mathbb{A}_K$, extend these linear maps to: $\mathbb{A}_E \rightarrow \mathbb{A}_K$. The topology on $\mathbb{A}_E$ is the coarsest topology for which all these extensions are continuous.

The topology can be defined in a different way. Fixing a basis for E over K results in an isomorphism $E \cong K^n$. Therefore fixing a basis induces an isomorphism $(\mathbb{A}_K)^n \cong \mathbb{A}_E$. The left-hand side is supplied with the product topology and transport this topology with the isomorphism onto the right-hand side. The topology doesn't depend on the choice of the basis, because another basis defines a second isomorphism. By composing both isomorphisms, a linear homeomorphism which transfers the two topologies into each other is obtained. More formally

$$\begin{aligned}
\mathbb{A}_E &= E \otimes_K \mathbb{A}_K \\
&\cong (K \otimes_K \mathbb{A}_K) \oplus \cdots \oplus (K \otimes_K \mathbb{A}_K) \\
&\cong \mathbb{A}_K \oplus \cdots \oplus \mathbb{A}_K
\end{aligned}$$

where the sums have n summands. In case of $E = L$, the definition above is consistent with the results about the adele ring of a finite extension L/K .

[9]

The adele ring of an algebra

Let A be a finite-dimensional algebra over K . In particular, A is a finite-dimensional vector-space over K . As a consequence, $\mathbb{A}_A$ is defined and $\mathbb{A}_A \cong \mathbb{A}_K \otimes_K A$. Since there is multiplication on $\mathbb{A}_K$ and A , a multiplication on $\mathbb{A}_A$ can be defined via:

$$\forall \alpha, \beta \in \mathbb{A}_K \text{ and } \forall a, b \in A : \quad (\alpha \otimes_K a) \cdot (\beta \otimes_K b) := (\alpha\beta) \otimes_K (ab).$$

As a consequence, $\mathbb{A}_A$ is an algebra with a unit over $\mathbb{A}_K$. Let B be a finite subset of A , containing a basis for A over K . For any finite place v , M_v is defined as the O_v -module generated by B in A_v . For each finite set of places, $P \supset P_\infty$, define

$$\mathbb{A}_A(P, \alpha) = \prod_{v \in P} A_v \times \prod_{v \notin P} M_v.$$

One can show there is a finite set P_0 , so that $\mathbb{A}_A(P, \alpha)$ is an open subring of $\mathbb{A}_A$, if $P \supset P_0$. Furthermore $\mathbb{A}_A$ is the union of all these subrings and for $A = K$, the definition above is consistent with the definition of the adele ring.

Trace and norm on the adele ring

Let L/K be a finite extension. Since $\mathbb{A}_K = \mathbb{A}_K \otimes_K K$ and $\mathbb{A}_L = \mathbb{A}_K \otimes_K L$ from the Lemma above, $\mathbb{A}_K$ can be interpreted as a closed subring of $\mathbb{A}_L$. For this embedding, write $\text{con}_{L/K}$. Explicitly for all places w of L above v and for any $\alpha \in \mathbb{A}_K$, $(\text{con}_{L/K}(\alpha))_w = \alpha_v \in K_v$.

Let $M/L/K$ be a tower of global fields. Then:

$$\text{con}_{M/K}(\alpha) = \text{con}_{M/L}(\text{con}_{L/K}(\alpha)) \quad \forall \alpha \in \mathbb{A}_K.$$

Furthermore, restricted to the principal adeles con is the natural injection $K \rightarrow L$.

Let $\{\omega_1, \dots, \omega_n\}$ be a basis of the field extension L/K . Then each $\alpha \in \mathbb{A}_L$ can be written as $\sum_{j=1}^n \alpha_j \omega_j$, where $\alpha_j \in \mathbb{A}_K$ are unique. The map $\alpha \mapsto \alpha_j$ is continuous. Define α_{ij} depending on α via the equations:

$$\begin{aligned}
\alpha \omega_1 &= \sum_{j=1}^n \alpha_{1j} \omega_j \\
&\vdots \\
\alpha \omega_n &= \sum_{j=1}^n \alpha_{nj} \omega_j
\end{aligned}$$

Now, define the trace and norm of α as:

$$\begin{aligned}\mathrm{Tr}_{L/K}(\alpha) &:= \mathrm{Tr}((\alpha_{ij})_{i,j}) = \sum_{i=1}^n \alpha_{ii} \\ N_{L/K}(\alpha) &:= N((\alpha_{ij})_{i,j}) = \det((\alpha_{ij})_{i,j})\end{aligned}$$

These are the trace and the determinant of the linear map

$$\begin{cases} \mathbb{A}_L \rightarrow \mathbb{A}_L \\ x \mapsto \alpha x \end{cases}$$

They are continuous maps on the adèle ring, and they fulfil the usual equations:

$$\begin{aligned}\mathrm{Tr}_{L/K}(\alpha + \beta) &= \mathrm{Tr}_{L/K}(\alpha) + \mathrm{Tr}_{L/K}(\beta) & \forall \alpha, \beta \in \mathbb{A}_L \\ \mathrm{Tr}_{L/K}(\mathrm{con}(\alpha)) &= n\alpha & \forall \alpha \in \mathbb{A}_K \\ N_{L/K}(\alpha\beta) &= N_{L/K}(\alpha)N_{L/K}(\beta) & \forall \alpha, \beta \in \mathbb{A}_L \\ N_{L/K}(\mathrm{con}(\alpha)) &= \alpha^n & \forall \alpha \in \mathbb{A}_K\end{aligned}$$

Furthermore, for $\alpha \in L$, $\mathrm{Tr}_{L/K}(\alpha)$ and $N_{L/K}(\alpha)$ are identical to the trace and norm of the field extension L/K . For a tower of fields $M/L/K$, the result is:

$$\begin{aligned}\mathrm{Tr}_{L/K}(\mathrm{Tr}_{M/L}(\alpha)) &= \mathrm{Tr}_{M/K}(\alpha) & \forall \alpha \in \mathbb{A}_M \\ N_{L/K}(N_{M/L}(\alpha)) &= N_{M/K}(\alpha) & \forall \alpha \in \mathbb{A}_M\end{aligned}$$

Moreover, it can be proven that:^[10]

$$\begin{aligned}\mathrm{Tr}_{L/K}(\alpha) &= \left(\sum_{w|v} \mathrm{Tr}_{L_w/K_v}(\alpha_w) \right)_v & \forall \alpha \in \mathbb{A}_L \\ N_{L/K}(\alpha) &= \left(\prod_{w|v} N_{L_w/K_v}(\alpha_w) \right)_v & \forall \alpha \in \mathbb{A}_L\end{aligned}$$

Properties of the adèle ring

Theorem.^[11] For every set of places S , $\mathbb{A}_{K,S}$ is a locally compact topological ring.

Remark. The result above also holds for the adèle ring of vector-spaces and algebras over K .

Theorem.^[12] K is discrete and cocompact in $\mathbb{A}_K$. In particular, K is closed in $\mathbb{A}_K$.

Proof. Prove the case $K = \mathbb{Q}$. To show $\mathbb{Q} \subset \mathbb{A}_{\mathbb{Q}}$ is discrete it is sufficient to show the existence of a neighbourhood of $\mathbf{0}$ which contains no other rational number. The general case follows via translation. Define

$$U := \left\{ (\alpha_p)_p \mid \forall p < \infty : |\alpha_p|_p \leq 1 \quad \text{and} \quad |\alpha_\infty|_\infty < 1 \right\} = \widehat{\mathbb{Z}} \times (-1, 1).$$

U is an open neighbourhood of $\mathbf{0} \in \mathbb{A}_{\mathbb{Q}}$. It is claimed that $U \cap \mathbb{Q} = \{0\}$. Let $\beta \in U \cap \mathbb{Q}$, then $\beta \in \mathbb{Q}$ and $|\beta|_p \leq 1$ for all p and therefore $\beta \in \mathbb{Z}$. Additionally, $\beta \in (-1, 1)$ and therefore $\beta = 0$. Next, to show compactness, define:

$$W := \left\{ (\alpha_p)_p \mid \forall p < \infty : |\alpha_p|_p \leq 1 \quad \text{and} \quad |\alpha_\infty|_\infty \leq \frac{1}{2} \right\} = \widehat{\mathbb{Z}} \times \left[-\frac{1}{2}, \frac{1}{2} \right].$$

Each element in $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ has a representative in W , that is for each $\alpha \in \mathbb{A}_{\mathbb{Q}}$, there exists $\beta \in \mathbb{Q}$ such that $\alpha - \beta \in W$. Let $\alpha = (\alpha_p)_p \in \mathbb{A}_{\mathbb{Q}}$, be arbitrary and p be a prime for which $|\alpha_p|_p > 1$. Then there exists $r_p = z_p/p^{x_p}$ with $z_p \in \mathbb{Z}$, $x_p \in \mathbb{N}$ and $|\alpha_p - r_p|_p \leq 1$. Replace α with $\alpha - r_p$ and let $q \neq p$ be another prime. Then:

$$|\alpha_q - r_p|_q \leq \max \left\{ |\alpha_q|_q, |r_p|_q \right\} \leq \max \left\{ |\alpha_q|_q, 1 \right\} \leq 1.$$

Next, it can be claimed that:

$$|\alpha_q - r_p|_q \leq 1 \iff |\alpha_q|_q \leq 1.$$

The reverse implication is trivially true. The implication is true, because the two terms of the strong triangle inequality are equal if the absolute values of both integers are different. As a consequence, the (finite) set of primes for which the components of α are not in $\mathbb{Z}_p$ is reduced by 1. With iteration, it can be deduced that there exists $r \in \mathbb{Q}$ such that $\alpha - r \in \widehat{\mathbb{Z}} \times \mathbb{R}$. Now select $s \in \mathbb{Z}$ such that $\alpha_\infty - r - s \in [-\frac{1}{2}, \frac{1}{2}]$. Then $\alpha - (r + s) \in W$. The continuous projection $\pi : W \rightarrow \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ is surjective, therefore $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$, as the continuous image of a compact set, is compact.

Corollary. Let E be a finite-dimensional vector-space over K . Then E is discrete and cocompact in $\mathbb{A}_E$.

Theorem. The following are assumed:

- $\mathbb{A}_{\mathbb{Q}} = \mathbb{Q} + \mathbb{A}_{\mathbb{Z}}$.
- $\mathbb{Z} = \mathbb{Q} \cap \mathbb{A}_{\mathbb{Z}}$.
- $\mathbb{A}_{\mathbb{Q}}/\mathbb{Z}$ is a divisible group.^[13]
- $\mathbb{Q} \subset \mathbb{A}_{\mathbb{Q}, \text{fin}}$ is dense.

Proof. The first two equations can be proved in an elementary way.

By definition $\mathbb{A}_{\mathbb{Q}}/\mathbb{Z}$ is divisible if for any $n \in \mathbb{N}$ and $y \in \mathbb{A}_{\mathbb{Q}}/\mathbb{Z}$ the equation $nx = y$ has a solution $x \in \mathbb{A}_{\mathbb{Q}}/\mathbb{Z}$. It is sufficient to show $\mathbb{A}_{\mathbb{Q}}$ is divisible but this is true since $\mathbb{A}_{\mathbb{Q}}$ is a field with positive characteristic in each coordinate.

For the last statement note that $\mathbb{A}_{\mathbb{Q}, \text{fin}} = \mathbb{Q}\widehat{\mathbb{Z}}$, because the finite number of denominators in the coordinates of the elements of $\mathbb{A}_{\mathbb{Q}, \text{fin}}$ can be reached through an element $q \in \mathbb{Q}$. As a consequence, it is sufficient to show $\mathbb{Z} \subset \widehat{\mathbb{Z}}$ is dense, that is each open subset $V \subset \widehat{\mathbb{Z}}$ contains an element of $\mathbb{Z}$. Without loss of generality, it can be assumed that

$$V = \prod_{p \in E} (a_p + p^{l_p} \mathbb{Z}_p) \times \prod_{p \notin E} \mathbb{Z}_p,$$

because $(p^m \mathbb{Z}_p)_{m \in \mathbb{N}}$ is a neighbourhood system of 0 in $\mathbb{Z}_p$. By Chinese Remainder Theorem there exists $l \in \mathbb{Z}$ such that $l \equiv a_p \pmod{p^{l_p}}$. Since powers of distinct primes are coprime, $l \in V$ follows.

Remark. $\mathbb{A}_{\mathbb{Q}}/\mathbb{Z}$ is not uniquely divisible. Let $y = (0, 0, \dots) + \mathbb{Z} \in \mathbb{A}_{\mathbb{Q}}/\mathbb{Z}$ and $n \geq 2$ be given. Then

$$\begin{aligned} x_1 &= (0, 0, \dots) + \mathbb{Z} \\ x_2 &= \left(\frac{1}{n}, \frac{1}{n}, \dots\right) + \mathbb{Z} \end{aligned}$$

both satisfy the equation $nx = y$ and clearly $x_1 \neq x_2$ (x_2 is well-defined, because only finitely many primes divide n). In this case, being uniquely divisible is equivalent to being torsion-free, which is not true for $\mathbb{A}_{\mathbb{Q}}/\mathbb{Z}$ since $nx_2 = 0$, but $x_2 \neq 0$ and $n \neq 0$.

Remark. The fourth statement is a special case of the strong approximation theorem.

Haar measure on the adèle ring

Definition. A function $f : \mathbb{A}_K \rightarrow \mathbb{C}$ is called simple if $f = \prod_v f_v$, where $f_v : K_v \rightarrow \mathbb{C}$ are measurable and $f_v = \mathbf{1}_{O_v}$ for almost all v .

Theorem.^[14] Since $\mathbb{A}_K$ is a locally compact group with addition, there is an additive Haar measure dx on $\mathbb{A}_K$. This measure can be normalised such that every integrable simple function $f = \prod_v f_v$ satisfies:

$$\int_{\mathbb{A}_K} f dx = \prod_v \int_{K_v} f_v dx_v,$$

where for $v < \infty$, dx_v is the measure on K_v such that O_v has unit measure and dx_∞ is the Lebesgue measure. The product is finite, i.e., almost all factors are equal to one.

The idele group

Definition. Define the **idele group of K** as the group of units of the adèle ring of K , that is $I_K := \mathbb{A}_K^\times$. The elements of the idele group are called the **ideles of K** .

Remark. I_K is equipped with a topology so that it becomes a topological group. The subset topology inherited from $\mathbb{A}_K$ is not a suitable candidate since the group of units of a topological ring equipped with subset topology may *not* be a topological group. For example, the inverse map in $\mathbb{A}_{\mathbb{Q}}$ is not continuous. The sequence

$$\begin{aligned}
x_1 &= (2, 1, \dots) \\
x_2 &= (1, 3, 1, \dots) \\
x_3 &= (1, 1, 5, 1, \dots) \\
&\vdots
\end{aligned}$$

converges to $1 \in \mathbb{A}_{\mathbb{Q}}$. To see this let U be neighbourhood of 0 , without loss of generality it can be assumed:

$$U = \prod_{p \leq N} U_p \times \prod_{p > N} \mathbb{Z}_p$$

Since $(x_n)_p - 1 \in \mathbb{Z}_p$ for all p , $x_n - 1 \in U$ for n large enough. However, as was seen above the inverse of this sequence does not converge in $\mathbb{A}_{\mathbb{Q}}$.

Lemma. Let R be a topological ring. Define:

$$\begin{cases} \iota : R^\times \rightarrow R \times R \\ x \mapsto (x, x^{-1}) \end{cases}$$

Equipped with the topology induced from the product on topology on $R \times R$ and ι , $R^\times$ is a topological group and the inclusion map $R^\times \subset R$ is continuous. It is the coarsest topology, emerging from the topology on R , that makes $R^\times$ a topological group.

Proof. Since R is a topological ring, it is sufficient to show that the inverse map is continuous. Let $U \subset R^\times$ be open, then $U \times U^{-1} \subset R \times R$ is open. It is necessary to show $U^{-1} \subset R^\times$ is open or equivalently, that $U^{-1} \times (U^{-1})^{-1} = U^{-1} \times U \subset R \times R$ is open. But this is the condition above.

The idele group is equipped with the topology defined in the Lemma making it a topological group.

Definition. For S a subset of places of K set: $I_{K,S} := \mathbb{A}_{K,S}^\times$, $I_K^S := (\mathbb{A}_K^S)^\times$.

Lemma. The following identities of topological groups hold:

$$\begin{aligned}
I_{K,S} &= \prod'_{v \in S} K_v^\times \\
I_K^S &= \prod'_{v \notin S} K_v^\times \\
I_K &= \prod'_v K_v^\times
\end{aligned}$$

where the restricted product has the restricted product topology, which is generated by restricted open rectangles of the form

$$\prod_{v \in E} U_v \times \prod_{v \notin E} O_v^\times,$$

where E is a finite subset of the set of all places and $U_v \subset K_v^\times$ are open sets.

Proof. Prove the identity for I_K ; the other two follow similarly. First show the two sets are equal:

$$\begin{aligned}
I_K &= \{x = (x_v)_v \in \mathbb{A}_K : \exists y = (y_v)_v \in \mathbb{A}_K : xy = 1\} \\
&= \{x = (x_v)_v \in \mathbb{A}_K : \exists y = (y_v)_v \in \mathbb{A}_K : x_v \cdot y_v = 1 \quad \forall v\} \\
&= \{x = (x_v)_v : x_v \in K_v^\times \forall v \text{ and } x_v \in O_v^\times \text{ for almost all } v\} \\
&= \prod'_v K_v^\times
\end{aligned}$$

In going from line 2 to 3, x as well as $x^{-1} = y$ have to be in $\mathbb{A}_K$, meaning $x_v \in O_v$ for almost all v and $x_v^{-1} \in O_v$ for almost all v . Therefore, $x_v \in O_v^\times$ for almost all v .

Now, it is possible to show the topology on the left-hand side equals the topology on the right-hand side. Obviously, every open restricted rectangle is open in the topology of the idele group. On the other hand, for a given $U \subset I_K$, which is open in the topology of the idele group, meaning $U \times U^{-1} \subset \mathbb{A}_K \times \mathbb{A}_K$ is open, so for each $u \in U$ there exists an open restricted rectangle, which is a subset of U and contains u . Therefore, U is the union of all these restricted open rectangles and therefore is open in the restricted product topology.

Lemma. For each set of places, S , $I_{K,S}$ is a locally compact topological group.

Proof. The local compactness follows from the description of $I_{K,S}$ as a restricted product. It being a topological group follows from the above discussion on the group of units of a topological ring.

A neighbourhood system of $1 \in \mathbb{A}_K(P_\infty)^\times$ is a neighbourhood system of $1 \in I_K$. Alternatively, take all sets of the form:

$$\prod_v U_v,$$

where U_v is a neighbourhood of $1 \in K_v^\times$ and $U_v = O_v^\times$ for almost all v .

Since the idele group is a locally compact, there exists a Haar measure $d^\times x$ on it. This can be normalised, so that

$$\int_{I_{K,\text{fin}}} \mathbf{1}_{\widehat{O}} d^\times x = 1.$$

This is the normalisation used for the finite places. In this equation, $I_{K,\text{fin}}$ is the finite idele group, meaning the group of units of the finite adele ring. For the infinite places, use the multiplicative lebesgue measure $\frac{dx}{|x|}$.

The idele group of a finite extension

Lemma. Let L/K be a finite extension. Then:

$$I_L = \prod'_v L_v^\times.$$

where the restricted product is with respect to $\widetilde{O}_v^\times$.

Lemma. There is a canonical embedding of I_K in I_L .

Proof. Map $a = (a_v)_v \in I_K$ to $a' = (a'_w)_w \in I_L$ with the property $a'_w = a_v \in K_v^\times \subset L_w^\times$ for $w|v$. Therefore, I_K can be seen as a subgroup of I_L . An element $a = (a_w)_w \in I_L$ is in this subgroup if and only if his components satisfy the following properties: $a_w \in K_v^\times$ for $w|v$ and $a_w = a_{w'}$ for $w|v$ and $w'|v$ for the same place v of K .

The case of vector spaces and algebras

[15]

The idele group of an algebra

Let A be a finite-dimensional algebra over K . Since $\mathbb{A}_A^\times$ is not a topological group with the subset-topology in general, equip $\mathbb{A}_A^\times$ with the topology similar to I_K above and call $\mathbb{A}_A^\times$ the idele group. The elements of the idele group are called idele of A .

Proposition. Let α be a finite subset of A , containing a basis of A over K . For each finite place v of K , let α_v be the O_v -module generated by α in A_v . There exists a finite set of places P_0 containing P_∞ such that for all $v \notin P_0$, α_v is a compact subring of A_v . Furthermore, α_v contains $A_v^\times$. For each v , $A_v^\times$ is an open subset of A_v and the map $x \mapsto x^{-1}$ is continuous on $A_v^\times$. As a consequence $x \mapsto (x, x^{-1})$ maps $A_v^\times$ homeomorphically on its image in $A_v \times A_v$. For each $v \notin P_0$, the $\alpha_v^\times$ are the elements of $A_v^\times$, mapping in $\alpha_v \times \alpha_v$ with the function above. Therefore, $\alpha_v^\times$ is an open and compact subgroup of $A_v^\times$. [16]

Alternative characterisation of the idele group

Proposition. Let $P \supset P_\infty$ be a finite set of places. Then

$$\mathbb{A}_A(P, \alpha)^\times := \prod_{v \in P} A_v^\times \times \prod_{v \notin P} \alpha_v^\times$$

is an open subgroup of $\mathbb{A}_A^\times$, where $\mathbb{A}_A^\times$ is the union of all $\mathbb{A}_A(P, \alpha)^\times$. [17]

Corollary. In the special case of $A = K$ for each finite set of places $P \supset P_\infty$,

$$\mathbb{A}_K(P)^\times = \prod_{v \in P} K_v^\times \times \prod_{v \notin P} O_v^\times$$

is an open subgroup of $\mathbb{A}_K^\times = I_K$. Furthermore, I_K is the union of all $\mathbb{A}_K(P)^\times$.

Norm on the idele group

The trace and the norm should be transfer from the adele ring to the idele group. It turns out the trace can't be transferred so easily. However, it is possible to transfer the norm from the adele ring to the idele group. Let $\alpha \in I_K$. Then $\text{con}_{L/K}(\alpha) \in I_L$ and therefore, it can be said that in injective group homomorphism

$$\text{con}_{L/K} : I_K \hookrightarrow I_L.$$

Since $\alpha \in I_L$, it is invertible, $N_{L/K}(\alpha)$ is invertible too, because $(N_{L/K}(\alpha))^{-1} = N_{L/K}(\alpha^{-1})$. Therefore $N_{L/K}(\alpha) \in I_K$. As a consequence, the restriction of the norm-function introduces a continuous function:

$$N_{L/K} : I_L \rightarrow I_K.$$

The Idele class group

Lemma. There is natural embedding of $K^\times$ into $I_{K,S}$ given by the diagonal map:
 $a \mapsto (a, a, a, \dots).$

Proof. Since $K^\times$ is a subset of $K_v^\times$ for all v , the embedding is well-defined and injective.

Corollary. $A^\times$ is a discrete subgroup of $\mathbb{A}_A^\times$.

Definition. In analogy to the ideal class group, the elements of $K^\times$ in I_K are called **principal ideles of I_K** . The quotient group $C_K := I_K/K^\times$ is called idele class group of K . This group is related to the ideal class group and is a central object in class field theory.

Remark. $K^\times$ is closed in I_K , therefore C_K is a locally compact topological group and a Hausdorff space.

Lemma.^[18] Let L/K be a finite extension. The embedding $\quad$ induces an injective map:

Properties of the idele group

Absolute value on the idele group of K and 1-idele

Definition. For $\alpha = (\alpha_v)$ define: $|\alpha| = \prod_v |\alpha_v|$. Since α is an idele this product is finite and therefore well-defined.

Remark. The definition can be extended to $\mathbb{A}_K$ by allowing infinite products. However, these infinite products vanish and so $|\cdot|$ vanishes on $\mathbb{A}_K$. $|\cdot|$ will be used to denote both the function on I_K and $\mathbb{A}_K$.

Theorem. $|\cdot|$ is a continuous group homomorphism.

Proof. Let

where it is used that all products are finite. The map is continuous which can be seen using an argument dealing with sequences. This reduces the problem to whether $|\cdot|$ is continuous on $\mathbb{A}_K$. However, this is clear, because of the reverse triangle inequality.

Definition. The set of $\mathbb{A}_K^\times$ -idele can be defined as:

I_K is a subgroup of $\mathbb{A}_K^\times$. Since $\mathbb{A}_K^\times$ is a closed subset of $\mathbb{A}_K$. Finally the $\mathbb{A}_K^\times$ -topology on I_K equals the subset-topology of I_K on $\mathbb{A}_K^\times$ [19][20]

Artin's Product Formula. $\prod_v \chi(v) = 1$ for all $\chi \in I_K^\vee$

Proof.[21] Proof of the formula for number fields, the case of global function fields can be proved similarly. Let K be a number field and $\chi \in I_K^\vee$. It has to be shown that:

For finite place v for which the corresponding prime ideal $\mathfrak{p}_v$ does not divide d_K , $\chi(v) = 1$ and therefore $\prod_v \chi(v) = 1$. This is valid for almost all v . There is:

In going from line 1 to line 2, the identity $\chi(v) = \chi(w)$ was used where v is a place of K and w is a place of $\mathbb{Q}$ lying above v . Going from line 2 to line 3, a property of the norm is used. The norm is in $\mathbb{Q}$ so without loss of generality it can be assumed that $N(\mathfrak{p}) = p$. Then $\mathfrak{p}$ possesses a unique integer factorisation:

where α_p is 0 for almost all p . By Ostrowski's theorem all absolute values on $\mathbb{Q}$ are equivalent to the real absolute value $|\cdot|_\infty$ or a p -adic absolute value. Therefore:

Lemma.^[22] There exists a constant c_K depending only on K , such that for every v satisfying $\|v\|_v \leq c_K$ there exists α_v such that $\alpha_v v$ is integral for all v .

Corollary. Let S be a place of K and let α_v be given for all $v \in S$ with the property $\alpha_v v$ is integral for almost all v . Then there exists α such that αv is integral for all $v \in S$.

Proof. Let c_K be the constant from the lemma. Let α_v be a uniformising element of $\mathcal{O}_v$. Define the adele α via $\alpha_v = \alpha_v \alpha_v^{-1}$ with α_v minimal, so that $\alpha_v v$ is integral for all $v \in S$. Then $\alpha_v v$ is integral for almost all v . Define α_v with $\alpha_v v$ integral for almost all v . This works, because $\alpha_v v$ is integral for almost all v . By the Lemma there exists α_v such that $\alpha_v v$ is integral for all $v \in S$.

Theorem. $K^\times$ is discrete and cocompact in $\mathbb{A}_K^\times$.

Proof.^[23] Since $K^\times$ is discrete in I_K it is also discrete in $\mathbb{A}_K^\times$. To prove the compactness of $K^\times \backslash \mathbb{A}_K^\times$ let c_K be the constant of the Lemma and suppose α_v satisfying $\alpha_v v$ is integral for almost all v is given. Define:

Clearly $\alpha_v v$ is compact. It can be claimed that the natural projection $\pi_v: \mathbb{A}_K^\times \rightarrow \mathbb{A}_K^\times / K^\times$ is surjective. Let α_v be arbitrary, then:

and therefore

It follows that

By the Lemma there exists α_v such that $\alpha_v v$ is integral for all v , and therefore $\alpha_v v$ is integral for almost all v , proving the surjectivity of the natural projection. Since it is also continuous the compactness follows.

Theorem.^[24] There is a canonical isomorphism $K^\times \backslash \mathbb{A}_K^\times \cong \prod_v \mathcal{O}_v^\times / \mathcal{O}_v^\times$. Furthermore, $\prod_v \mathcal{O}_v^\times / \mathcal{O}_v^\times$ is a set of representatives for $K^\times \backslash \mathbb{A}_K^\times$ and $\prod_v \mathcal{O}_v^\times / \mathcal{O}_v^\times$ is a set of representatives for $K^\times \backslash \mathbb{A}_K^\times$.

Proof. Consider the map

This map is well-defined, since $\phi(a/b) = \phi(a)/\phi(b)$ for all p and therefore ϕ is a continuous group homomorphism. Now suppose $\phi(a/b) = 1$. Then there exists n such that $a/b = 1 + n$. By considering the infinite place it can be seen that n proves injectivity. To show surjectivity let $\alpha \in K^\times$. The absolute value of this element is $|\alpha|$ and therefore

Hence ϕ and there is:

Since

It has been concluded that ϕ is surjective.

Theorem.^[24] The absolute value function induces the following isomorphisms of topological groups:

Proof. The isomorphisms are given by:

Relation between ideal class group and idele class group

Theorem. Let K be a number field with ring of integers $\mathcal{O}_K$, group of fractional ideals I_K and ideal class group Cl_K . Here's the following isomorphisms

where φ_v has been defined.

Proof. Let $\mathfrak{p}$ be a finite place of K and let α be a representative of the equivalence class $\mathfrak{p}$. Define

Then $\mathfrak{p}$ is a prime ideal in $\mathcal{O}_K$. The map φ_v is a bijection between finite places of K and non-zero prime ideals of $\mathcal{O}_K$. The inverse is given as follows: a prime ideal $\mathfrak{p}$ is mapped to the valuation $v_{\mathfrak{p}}$ given by

The following map is well-defined:

The map φ is obviously a surjective homomorphism and $\varphi(\mathcal{O}_K^\times) = \mathcal{O}_K^\times$. The first isomorphism follows from fundamental theorem on homomorphism. Now, both sides are divided by $\mathcal{O}_K^\times$. This is possible, because

Please, note the abuse of notation: On the left hand side in line 1 of this chain of equations, φ stands for the map defined above. Later, the embedding of $K^\times$ into $I_{K,\mathfrak{m}}$ is used. In line 2, the definition of the map is used. Finally, use that $\mathcal{O}_K$ is a Dedekind domain and therefore each ideal can be written as a product of prime ideals. In other words, the map φ is a $K^\times$ -equivariant group homomorphism. As a consequence, the map above induces a surjective homomorphism

To prove the second isomorphism, it has to be shown that φ is surjective. Consider $v \in \mathbb{A}_K^\times$. Then $v = \prod_{\mathfrak{p}} v_{\mathfrak{p}}$ because $v_{\mathfrak{p}} = 1$ for all but finitely many $\mathfrak{p}$. On the other hand, consider $v_{\mathfrak{p}}$ with $v_{\mathfrak{p}} \neq 1$. Then $v_{\mathfrak{p}} = u_{\mathfrak{p}} \pi_{\mathfrak{p}}^{n_{\mathfrak{p}}}$ where $u_{\mathfrak{p}} \in \mathcal{O}_{\mathfrak{p}}^\times$ and $\pi_{\mathfrak{p}}$ is a uniformizer. Then $u_{\mathfrak{p}} = 1 + \pi_{\mathfrak{p}} w_{\mathfrak{p}}$ for some $w_{\mathfrak{p}} \in \mathcal{O}_{\mathfrak{p}}$. Then $u_{\mathfrak{p}} = 1 + \pi_{\mathfrak{p}} w_{\mathfrak{p}}$ which allows to write $u_{\mathfrak{p}} = 1 + \pi_{\mathfrak{p}} w_{\mathfrak{p}}$. As a consequence, there exists a representative, such that: $u_{\mathfrak{p}} = 1 + \pi_{\mathfrak{p}} w_{\mathfrak{p}}$. Consequently, $v = \prod_{\mathfrak{p}} u_{\mathfrak{p}} \pi_{\mathfrak{p}}^{n_{\mathfrak{p}}}$ and therefore $v = u \prod_{\mathfrak{p}} \pi_{\mathfrak{p}}^{n_{\mathfrak{p}}}$ where $u = \prod_{\mathfrak{p}} u_{\mathfrak{p}}$. The second isomorphism of the theorem has been proven.

For the last isomorphism note that φ induces a surjective group homomorphism $\varphi: \mathbb{A}_K^\times \rightarrow K^\times$ with $\ker \varphi = \mathbb{A}_K^\times$.

Remark. Consider $I_{K, \text{fin}}$ with the idele topology and equip $K^\times$ with the discrete topology. Since $\mathcal{O}_{\mathfrak{p}}^\times$ is open for each $\mathfrak{p}$, φ is continuous. It stands, that φ is open, where φ is surjective so that φ is an isomorphism.

Decomposition of the idele group and idele class group of K

Theorem.

Proof. For each place v of K let $\mathcal{O}_v$ be the local ring of integers. Then $\mathcal{O}_v^\times$ is the group of units of $\mathcal{O}_v$. For each place v of K let π_v be a uniformizer. Then $\mathcal{O}_v^\times = \langle \pi_v \rangle \times \mathcal{O}_v^\times$ so that for all $x \in \mathcal{O}_v^\times$ there exists a unique $n_v \in \mathbb{Z}$ and $u_v \in \mathcal{O}_v^\times$ such that $x = \pi_v^{n_v} u_v$. Then $x = \pi_v^{n_v} u_v$ belongs to the subgroup of $\mathcal{O}_v^\times$ generated by π_v . Therefore for each $x \in \mathbb{A}_K^\times$ there exists a unique $n_v \in \mathbb{Z}$ and $u_v \in \mathcal{O}_v^\times$ such that $x_v = \pi_v^{n_v} u_v$. Then $x_v = \pi_v^{n_v} u_v$ is in the subgroup of $\mathcal{O}_v^\times$ generated by π_v . Therefore the image of the homomorphism φ is a discrete subgroup of $K^\times$ generated by π_v . Since this group is non-trivial, it is generated by π_v for some v . Choose v so that $\pi_v \in K^\times$ then I_K is the direct product of $\langle \pi_v \rangle$ and the subgroup generated by $\mathcal{O}_v^\times$. This subgroup is discrete and isomorphic to $\mathbb{Z}$.

For $v \neq v$ define:

The map $\varphi_v: \mathcal{O}_v^\times \rightarrow K^\times$ is an isomorphism of $\mathcal{O}_v^\times$ in a closed subgroup M of I_K and φ_v . The isomorphism is given by multiplication:

Obviously, φ_v is a homomorphism. To show it is injective, let $x \in \mathcal{O}_v^\times$ such that $\varphi_v(x) = 1$. Since $x = \pi_v^{n_v} u_v$ for $n_v \in \mathbb{Z}$ and $u_v \in \mathcal{O}_v^\times$ it stands that $\pi_v^{n_v} u_v = 1$ for $v < \infty$. Moreover, it exists a $n_v \in \mathbb{Z}$ so that $\pi_v^{n_v} = 1$ for $v | \infty$. Therefore, $\pi_v^{n_v} = 1$ for $v | \infty$. Moreover $u_v = 1$ implies $x = 1$ where n is the number of infinite places of K . As a consequence φ_v is injective. To show surjectivity, let $x \in K^\times$. It is defined that $x = \pi_v^{n_v} u_v$ and furthermore, $\pi_v^{n_v} = 1$ for $v < \infty$ and $u_v = 1$ for $v | \infty$. Define $x = \prod_{\mathfrak{p}} \pi_{\mathfrak{p}}^{n_{\mathfrak{p}}} u_{\mathfrak{p}}$. It stands, that $x \in I_K$. Therefore, φ is surjective.

The other equations follow similarly.

Characterisation of the idele group

Theorem.^[25] Let K be a number field. There exists a finite set of places S such that:

Proof. The class number of a number field is finite so let $\mathcal{C}$ be the set of ideal classes. Let $\mathfrak{a}_1, \dots, \mathfrak{a}_h$ be the ideals, representing the classes in $\mathcal{C}$. These ideals are generated by a finite number of prime ideals. Let S be a finite set of places containing P_∞ and the finite places corresponding to $\mathfrak{a}_1, \dots, \mathfrak{a}_h$. Consider the isomorphism:

induced by

At infinite places the statement is immediate, so the statement has been proved for finite places. The inclusion " $\supseteq$ " is obvious. Let $\mathfrak{a}$ be an ideal. The corresponding ideal $\mathfrak{a}$ belongs to a class $\mathfrak{a}\mathcal{C}$ meaning $\mathfrak{a} = \alpha \mathfrak{b}$ for a principal ideal $\mathfrak{b}$. The idele α maps to the ideal $\mathfrak{b}$ under the map $\alpha \mapsto \mathfrak{b}$. That means $\alpha \mathfrak{a} = \mathfrak{b} \mathfrak{a}$. Since the prime ideals in S are in $\mathfrak{a}$ it follows that $\alpha \mathfrak{a} = \mathfrak{b} \mathfrak{a}$ for all $\mathfrak{a}$ that means $\alpha = \mathfrak{b}$ for all $\mathfrak{a}$. It follows, that therefore

Applications

Finiteness of the class number of a number field

In the previous section the fact that the class number of a number field is finite had been used. Here this statement can be proved:

Theorem (finiteness of the class number of a number field). Let K be a number field. Then

Proof. The map

is surjective and therefore $\pi^{-1}(S)$ is the continuous image of the compact set S . Thus, $\pi^{-1}(S)$ is compact. In addition, it is discrete and so finite.

Remark. There is a similar result for the case of a global function field. In this case, the so-called divisor group is defined. It can be shown that the quotient of the set of all divisors of degree 0 by the set of the principal divisors is a finite group.^[26]

Group of units and Dirichlet's unit theorem

Let $P \supset P_\infty$ be a finite set of places. Define

Then $\mathcal{O}_P^\times$ is a subgroup of $K^\times$ containing all elements satisfying $|x|_v \leq 1$ for all $v \in P$. Since $K^\times$ is discrete in I_K , $\mathcal{O}_P^\times$ is a discrete subgroup of I_K and with the same argument, $\mathcal{O}_P^\times$ is discrete in I_K .

An alternative definition is: $\mathcal{O}_P^\times$ where $\mathcal{O}_P$ is a subring of K defined by

As a consequence, $\mathcal{O}_P^\times$ contains all elements $x \in K^\times$ which fulfil $|x|_v \leq 1$ for all $v \in P$.

Lemma 1. Let S be a finite set of places. The following set is finite:

Proof. Define

$\mathcal{O}_S^\times$ is compact and the set described above is the intersection of $\mathcal{O}_S^\times$ with the discrete subgroup $K^\times$ in I_K and therefore finite.

Lemma 2. Let E be set of all $x \in K^\times$ such that $|x|_v = 1$ for all $v \in P$. Then E is the group of all roots of unity of K . In particular it is finite and cyclic.

Proof. All roots of unity of K have absolute value 1 so E contains all roots of unity. For converse note that Lemma 1 with $S = P$ implies E is finite. Moreover $|x|_v = 1$ for each finite set of places P . Finally suppose there exists $x \in E$ which is not a root of unity of K . Then $|x|_v \neq 1$ for all $v \in P$ contradicting the finiteness of E .

Unit Theorem. $\mathcal{O}_P^\times$ is the direct product of E and a group isomorphic to $\mathbb{Z}^{r-1}$ where r is the rank of the free part of the class group and $r = r_1 + r_2$ if K is a real quadratic field and $r = r_1$ if K is a complex quadratic field.^[27]

Dirichlet's Unit Theorem. Let K be a number field. Then
where G is the finite cyclic group of all roots of unity
of K is the number of real embeddings of K and r_2 is the number of conjugate
pairs of complex embeddings of K . It stands, that

Remark. The Unit Theorem generalises Dirichlet's Unit Theorem. To see this, let K be a number field. It is already known that
set S and note

Then there is:

Approximation theorems

Weak Approximation Theorem.^[28] Let S be inequivalent
valuations of K . Let K_S be the completion of K with respect to S . Embed K
diagonally in $\prod_{v \in S} K_v$. Then K is everywhere dense in
In other words, for each $(a_v)_{v \in S}$ and for each $\epsilon > 0$
there exists $x \in K$ such that:

Strong Approximation Theorem.^[29] Let v be a place of K . Define

Then K is dense in

Remark. The global field is discrete in its adele ring. The strong approximation theorem tells us that, if one place (or more) is
omitted, the property of discreteness of K is turned into a denseness of K .

Hasse principle

Hasse-Minkowski Theorem. A quadratic form on K is zero, if and only if, the
quadratic form is zero in each completion

Remark. This is the Hasse principle for quadratic forms. For polynomials of degree larger than 2 the Hasse principle isn't valid
in general. The idea of the Hasse principle (also known as local–global principle) is to solve a given problem of a number field K
by doing so in its completions K_v and then concluding on a solution in K .

Characters on the adèle ring

Definition. Let G be a locally compact abelian group. The character group of G is the set of all characters of G and is denoted by $G^\vee$. Equivalently $G^\vee$ is the set of all continuous group homomorphisms from G to $\mathbb{C}^\times$. Equip $G^\vee$ with the topology of uniform convergence on compact subsets of G . One can show that $G^\vee$ is also a locally compact abelian group.

Theorem. The adèle ring is *self-dual*:

Proof. By reduction to local coordinates, it is sufficient to show each K_v is self-dual. This can be done by using a fixed character of K_v . The idea has been illustrated by showing $\mathbb{R}$ is self-dual. Define:

Then the following map is an isomorphism which respects topologies:

Theorem (algebraic and continuous duals of the adèle ring).^[30] Let χ be a non-trivial character of $\mathbb{A}_K$, which is trivial on K . Let E be a finite-dimensional vector-space over K . Let $E^\vee$ and $\mathbb{A}_E$ be the algebraic duals of E and $\mathbb{A}_E$. Denote the topological dual of $\mathbb{A}_E$ by $\mathbb{A}_E^\vee$ and use $\langle \cdot, \cdot \rangle_E$ and $\langle \cdot, \cdot \rangle_{\mathbb{A}_E}$ to indicate the natural bilinear pairings on $E \times E^\vee$ and $\mathbb{A}_E \times \mathbb{A}_E^\vee$. Then the formula

$$\langle x, \chi \rangle = \langle x, \chi|_E \rangle_E$$

for all $x \in E$ determines an isomorphism $\mathbb{A}_E^\vee \xrightarrow{\sim} \mathbb{A}_E$ of $\mathbb{A}_E^\vee$ onto $\mathbb{A}_E$ where $\chi|_E$ and $\langle \cdot, \cdot \rangle_E$ are as above. Moreover, if $\chi|_K$ fulfils

$$\langle x, \chi|_K \rangle = \langle x, \chi|_K \rangle_K$$

for all $x \in K$ then

Tate's thesis

With the help of the characters of $\mathbb{A}_K$, Fourier analysis can be done on the adèle ring.^[31] John Tate in his thesis "Fourier analysis in Number Fields and Hecke Zeta Functions"^[6] proved results about Dirichlet L-functions using Fourier analysis on the adèle ring and the idele group. Therefore, the adèle ring and the idele group have been applied to study the Riemann zeta function and more general zeta functions and the L-functions. Adelic forms of these functions can be defined and represented as integrals over the adèle ring or the idele group, with respect to corresponding Haar measures. Functional equations and meromorphic continuations of these functions can be shown. For example, for all $s \in \mathbb{C}$ with

where $d^\times x$ is the unique Haar measure on $\mathbb{A}_K^\times$ normalised such that $\int_{\mathbb{A}_K^\times} d^\times x$ has volume one and is extended by zero to the finite adèle ring. As a result, the Riemann zeta function can be written as an integral over (a subset of) the adèle ring.^[32]

Automorphic forms

The theory of automorphic forms is a generalisation of Tate's thesis by replacing the idele group with analogous higher dimensional groups. To see this note:

Based on these identification a natural generalisation would be to replace the idele group and the 1-idele with:

And finally

where $\mathbb{A}_f$ is the centre of $\mathbb{A}_f$. Then it define an automorphic form as an element of $\mathbb{A}_f^\times$. In other words an automorphic form is a function on $\mathbb{A}_f^\times$ satisfying certain algebraic and analytic conditions. For studying automorphic forms, it is important to know the representations of the group $\mathrm{GL}(n, \mathbb{A}_f)$. It is also possible to study automorphic L-functions, which can be described as integrals over $\mathbb{A}_f^\times$ [33]

Generalise even further is possible by replacing $\mathbb{Q}$ with a number field and $\mathrm{GL}(n, \mathbb{A}_f)$ with an arbitrary reductive algebraic group.

Further applications

A generalisation of Artin reciprocity law leads to the connection of representations of $\mathrm{GL}(n, \mathbb{A}_f)$ and of Galois representations of K (Langlands program).

The idele class group is a key object of class field theory, which describes abelian extensions of the field. The product of the local reciprocity maps in local class field theory gives a homomorphism of the idele group to the Galois group of the maximal abelian extension of the global field. The Artin reciprocity law, which is a sweeping generalisation of the Gauss quadratic reciprocity law, states that the product vanishes on the multiplicative group of the number field. Thus, the global reciprocity map of the idele class group to the abelian part of the absolute Galois group of the field will be obtained.

The self-duality of the adèle ring of the function field of a curve over a finite field easily implies the Riemann–Roch theorem and the duality theory for the curve.

References

1. Groechenig, Michael (August 2017). "Adelic Descent Theory". *Compositio Mathematica*. **153** (8): 1706–1746. arXiv:1511.06271 (https://arxiv.org/abs/1511.06271). doi:10.1112/S0010437X17007217 (https://doi.org/10.1112%2FS0010437X17007217). ISSN 0010-437X (https://search.worldcat.org/issn/0010-437X). S2CID 54016389 (https://api.semanticscholar.org/CorpusID:54016389).
2. Sutherland, Andrew (1 December 2015). *18.785 Number theory I Lecture #22* (https://math.mit.edu/classes/18.785/2015fa/LectureNotes22.pdf) (PDF). MIT. p. 4.
3. "ring of adeles in nLab" (https://ncatlab.org/nlab/show/ring+of+adeles). *ncatlab.org*.
4. *Geometric Class Field Theory, notes by Tony Feng of a lecture of Bhargav Bhatt* (https://math.berkeley.edu/~fengt/2GeometricCFT.pdf) (PDF).

5. *Weil uniformization theorem, nlab article* (<https://ncatlab.org/nlab/show/Weil+uniformization+theorem>).
6. Cassels & Fröhlich 1967.
7. Tate, John (1968), "Residues of differentials on curves" ([http://archive.numdam.org/ARCHIVE/ASENS/ASENS_1968_4_1_1_149_0/ASENS_1968_4_1_1_149_0.pdf](http://archive.numdam.org/ARCHIVE/ASENS/ASENS_1968_4_1_1/ASENS_1968_4_1_1_149_0/ASENS_1968_4_1_1_149_0.pdf)) (PDF), *Annales Scientifiques de l'École Normale Supérieure*, 1: 149–159, doi:10.24033/asens.1162 (<https://doi.org/10.24033%2Fasens.1162>).
8. This proof can be found in Cassels & Fröhlich 1967, p. 64.
9. The definitions are based on Weil 1967, p. 60.
10. See Weil 1967, p. 64 or Cassels & Fröhlich 1967, p. 74.
11. For proof see Deitmar 2010, p. 124, theorem 5.2.1.
12. See Cassels & Fröhlich 1967, p. 64, Theorem, or Weil 1967, p. 64, Theorem 2.
13. The next statement can be found in Neukirch 2007, p. 383.
14. See Deitmar 2010, p. 126, Theorem 5.2.2 for the rational case.
15. This section is based on Weil 1967, p. 71.
16. A proof of this statement can be found in Weil 1967, p. 71.
17. A proof of this statement can be found in Weil 1967, p. 72.
18. For a proof see Neukirch 2007, p. 388.
19. This statement can be found in Cassels & Fröhlich 1967, p. 69.
20. is also used for the set of the -idele but is used in this example.
21. There are many proofs for this result. The one shown below is based on Neukirch 2007, p. 195.
22. For a proof see Cassels & Fröhlich 1967, p. 66.
23. This proof can be found in Weil 1967, p. 76 or in Cassels & Fröhlich 1967, p. 70.
24. Part of Theorem 5.3.3 in Deitmar 2010.
25. The general proof of this theorem for any global field is given in Weil 1967, p. 77.
26. For more information, see Cassels & Fröhlich 1967, p. 71.
27. A proof can be found in Weil 1967, p. 78 or in Cassels & Fröhlich 1967, p. 72.
28. A proof can be found in Cassels & Fröhlich 1967, p. 48.
29. A proof can be found in Cassels & Fröhlich 1967, p. 67
30. A proof can be found in Weil 1967, p. 66.
31. For more see Deitmar 2010, p. 129.
32. A proof can be found Deitmar 2010, p. 128, Theorem 5.3.4. See also p. 139 for more information on Tate's thesis.
33. For further information see Chapters 7 and 8 in Deitmar 2010.

Sources

- Cassels, John; Fröhlich, Albrecht (1967). *Algebraic number theory: proceedings of an instructional conference, organized by the London Mathematical Society, (a NATO Advanced Study Institute)*. Vol. XVIII. London: Academic Press. ISBN 978-0-12-163251-9. 366 pages.
- Neukirch, Jürgen (2007). *Algebraische Zahlentheorie, unveränd. nachdruck der 1. Aufl. edn* (in German). Vol. XIII. Berlin: Springer. ISBN 9783540375470. 595 pages.
- Weil, André (1967). *Basic number theory*. Vol. XVIII. Berlin; Heidelberg; New York: Springer. ISBN 978-3-662-00048-9. 294 pages.
- Deitmar, Anton (2010). *Automorphe Formen* (in German). Vol. VIII. Berlin; Heidelberg (u.a.): Springer. ISBN 978-3-642-12389-4. 250 pages.
- Lang, Serge (1994). *Algebraic number theory, Graduate Texts in Mathematics 110* (2nd ed.). New York: Springer-Verlag. ISBN 978-0-387-94225-4.

External links

- What problem do the adeles solve? (<https://mathoverflow.net/questions/78240/what-problem-do-the-adeles-solve>)
- Some good books on adeles (<https://mathoverflow.net/questions/184737/a-good-book-on-adeles-and-ideles>)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Adele_ring&oldid=1303987058"